

A Data-Driven Framework for Performance Benchmarking in Football Using Normalized Expected Goals (xG) Metrics

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Abstract— In this paper, we are presenting our contribution to performance evaluation in Moroccan football by proposing a specific normalization method for existing xG-derived metrics. While the use of Expected Goals (xG) and its derivatives is well established in football analytics, our work focuses on adapting and standardizing these metrics to better reflect player, goalkeeper, team, and coaching performance within the local context.

By applying an XGBoost model to a dataset of shots collected from StatsBomb, we estimated xG values using a selected set of key variables. From these predictions, we then calculated several xG-derived metrics, which we normalized to better capture differences in player, goalkeeper, and team performance across various match situations.

These metrics demonstrate how players and teams participate in the game. They are useful tools for comparing performance, understanding team tactics, and helping players improve. The results of our study show that these indicators can provide helpful information for coaches and clubs, and support better decisions based on data in Moroccan football.

Keywords— Expected Goals (xG), Machine Learning, Football Analytics, Performance Metrics, Tactical Analysis, Player Evaluation, Goalkeeper Performance, Team Performance

I. INTRODUCTION

In recent years, performance analysis has become a crucial aspect of modern football, providing teams with valuable insights to gain competitive advantages [9]. Among the various advanced metrics used in football analytics, Expected Goals (xG) has emerged as a widely accepted quantitative measure to assess the quality of scoring opportunities [1]. The xG metric estimates the probability of a shot resulting in a goal based on factors such as shot location, angle, shot type, and defensive pressure.

While xG and related metrics are well-established in football analytics, they often have limitations. These include aggregating raw probabilities without considering contextual factors like playing time, shoot volume, or player roles. Such aggregation can lead to biased evaluations, potentially overestimating the effectiveness of players with more playing time or shots, or misrepresenting goalkeeper performance when shot difficulty is not properly considered. Additionally, traditional xG-derived generally lack normalization procedures for fair comparisons across players, goalkeepers, teams, and coaches.

Building on these foundations, this paper proposes a specific normalization method for existing xG-derived metrics

adjusted to the Moroccan football context. We apply an XGBoost model trained on a dataset of 5,213 shots from the Botola Pro 1 2021-2022 season to calculate xG values. We evaluated the model's performance using ROC-AUC, recall, and precision to ensure accurate and reliable predictions. From these predictions, we calculate normalized xG-derived indicators that we will be used to evaluate the performance of players, goalkeepers, teams, and coaches.

Our approach aims to provide an equitable evaluation of players, goalkeepers, teams, and coaches in Moroccan football, reflecting their true contributions in various match situations. By adapting and standardizing these metrics, we seek to better capture performance differences across different roles and contexts within the local football landscape.

The remainder of this paper is organized as follows: a review of relevant literature, a proposed normalization of xG-Derived metrics, a detailed explanation of our methodology, presentation and discussion of results, and concluding remarks with future research directions.

II. RELATED WORKS

Modern football analysis has been significantly transformed by data-driven metrics, particularly Expected Goals (xG) and their derivatives. This section provides a comprehensive review of academic literature on the application of xG and its metrics derived in football performance evaluation.

Research has explored several avenues for xG normalization. For team evaluation, adjusting xG based on opponent strength has been proposed as an effective method for a more reliable performance indicator by Ruan et al. [3]. Furthermore, Mead, O'Hare, and McMenemy [9] highlight that the effectiveness of xG models can be significantly improved by incorporating contextual factors such as player/team ability and psychological effects, demonstrating xG's superior predictive ability for a football team's future success compared to traditional statistics. For players, the most common normalization is the calculation of xG per 90 minutes of play (xG/90), which allows for a fair comparison of players with different playing times, as discussed by Markopoulou et al. [4]. Additionally, more sophisticated models, like Bayesian hierarchical models, have been employed to account for inter-player variability and the influence of their position on the field, offering a more individualized assessment, as demonstrated by Iapteff et al. [5].

Goalkeeper evaluation has necessitated the development of specific derived metrics. The Expected Goals on Target

(xGOT or PSxG) model not only assesses the quality of the chance (xG) but also considers the quality of the shot itself (e.g., its placement in the goal). This allows for a fairer evaluation of goalkeeper performance by separating it from the quality of the shots they face, as detailed by De-la-Cruz-Torres et al. [7] and Fu [6]. Consequently, these xGOT models represent a form of implicit normalization, as they evaluate the goalkeeper's ability relative to the difficulty of the action they must perform.

However, while these contributions provide valuable insights, they remain mostly descriptive and fragmented. Many frameworks address either players, teams, or goalkeepers separately, and only a few attempt to ensure consistency across roles. In addition, most of the existing works are based on European leagues, where tactical structures, data availability, and competition levels differ from those in developing leagues. This creates a gap in applicability when analyzing contexts such as Moroccan football, where playing styles and resources are different.

Despite these advancements, the literature presents notable gaps. Firstly, there is no unified normalization framework that consistently applies to the evaluation of players, goalkeepers, teams, and, most importantly, coaches. Our work aims to address this gap by proposing an xG-derived normalization methodology that adapts xG metrics to local contexts while keeping comparability with international approaches. The following section will detail the methodology developed to achieve this objective.

III. PROPOSED xG-DERIVED METRICS

A. Player Evaluation Metrics

To assess individual player contributions beyond conventional metrics such as goals and assists, we introduce other xG-derived metrics that offer a more granular evaluation of performance.

- **xG Player:** This metric calculates an xG normalized of a player, it quantifies a player's overall involvement in generating scoring opportunities, calculated as the sum of xG values for all shots taken by the player. We will use this formula

$$xG_{\text{Player}} = \frac{\sum_{i=1}^N xG_i}{N} \quad (1)$$

Where xG_i is the value of Expected Goals for the i th shot, and N denotes the total shots attempted by the player.

- **xG Efficiency:** This metric measures a player's ability to convert scoring opportunities into goals, calculated as the ratio of goals scored to the player's total xG. For this we will use this formula

$$xG \text{ Efficiency} = \left(\frac{\sum_{i=1}^N \text{Goals}}{N} - xG_{\text{player}} \right) \quad (2)$$

B. Goalkeeper Evaluation Metrics

For goalkeepers, Expected Goals Prevented provides a measure of a goalkeeper's shot-stopping ability, and exceeding expectations in preventing goals is considered as a testament to their skill.

$$xGP = \frac{\sum_{i=1}^N xG_{\text{allowed},i}}{\text{Goal_Conceded} + \sum_{i=1}^N xG_{\text{Allowed},i}} \quad (3)$$

where $xG_{\text{allowed},i}$ is the expected goals value of the i th shot that was saved, N denotes the total number of shots saved, and $\text{Goals}_{\text{conceded}}$ is the number of goals conceded.

C. Team Evaluation Metrics

Evaluating team performance necessitates a comprehensive assessment of both attacking and defensive capabilities, extending beyond simple goal tallies to encompass the underlying processes that drive success.

- **Net xG Difference:** this metric Indicates a team's overall dominance over its opponents. The formula is:

$$\text{NetxG} = \frac{1}{N} \sum_{i=1}^N xG_{\text{offensive},i} - \frac{1}{M} \sum_{j=1}^M xG_{\text{defensive},j} \quad (4)$$

where N is the number of shots generated, M is the number of shots conceded, $xG_{\text{offensive},i}$ is the expected goals value of the i th shot created, and $xG_{\text{defensive},j}$ is the expected goals value of the j th shot conceded.

- **Team xG Efficiency:** This metric measures a team's overall offensive effectiveness. The formula is:

$$\text{TeamxG} = \frac{\text{Goals} - \sum_{i=1}^N xG_{\text{offensive},i}}{N} \quad (5)$$

- **Defensive xG Efficiency:** This metric evaluates defensive performance. indicating whether the team concedes less than expected. The formula is:

$$\text{DefensivexGEff} = \frac{\sum_{j=1}^M xG_{\text{defensive},j} - \text{Goals}}{M} \quad (6)$$

D. Coach Evaluation Metrics

Evaluating coach performance poses a unique challenge, as it requires assessing the impact of tactical decisions and player management on overall team performance.

- **Opponent Adjusted xG:** This metric allows you to compare the team's performance against opponents of different levels, the formula is:

$$OAxG = xG_{\text{offensive},i} \times \left(1 - \frac{1}{M_{\text{adv}}} \sum_{j=1}^{M_{\text{adv}}} xG_{\text{defensive},j} \right) \quad (7)$$

where M_{adv} is the number of shots conceded by the opposing team in this season before the match.

- **xG Time Adjustment (xGTA):** This metric assesses the coach's ability to manage the team in different phases of the match, the formula is:

$$xGTA(\text{team}, T) = \sum_{i=1}^{NT} xG_{\text{generated},i} - \sum_{j=1}^{MT} xG_{\text{conceded},j} \quad (8)$$

where NT is the number of shots generated by the team during the time period T , $xG_{\text{generated},i}$ is the expected goals value of the i th generated shot, MT is the number of shots conceded during the same period, and $xG_{\text{conceded},j}$ is the expected goals value of the j th conceded shot.

The time interval T can take values in $[0-15]$, $[15-30]$, $[30-45]$, $[45-60]$, $[60-75]$, $[75-90]$ minutes.

The choice of metrics was motivated by the need for fairness and practical use. For players, efficiency indicators reveal finishing quality beyond raw goal counts. For goalkeepers, xGP captures shot-stopping relative to difficulty, which is more realistic than simple saves. For teams and coaches, metrics such as Net xG and xGTA provide a balanced view of attacking and defensive processes. These indicators are not perfect, since they depend on shot data and do not capture all tactical details, but they offer a practical and interpretable way to evaluate performance in real match conditions.

Having defined the proposed xG-derived metrics for evaluating player, goalkeeper, team, and coach performance, the subsequent section outlines the comprehensive methodology employed to develop and validate these proposed indicators.

IV. METHODOLOGY

To develop and validate the proposed xG-derived metrics, a comprehensive methodology was implemented, involving data collection, feature selection, model training, and performance evaluation.

A. Data collection and processing

The data used for this research is event data from StatsBomb, the dataset comprising 71,976 occurrences, for season 2021-2022 of the Botola Pro 1 (Moroccan professional football league).

The raw match event data underwent rigorous preprocessing to ensure data quality and consistency, a critical step in building robust and reliable xG models. The resulting data has 5,213 shots from all the season, with 30 columns of information for each shot.

B. Feature Selection

Feature selection was performed to extract relevant features from the raw data, such as distance to goal, shot angle, distance and angle to the nearest defenders, distance and angle to goalkeeper, and the body part used for the shot.

TABLE I. SUMMARY STATISTICS FOR PROPOSED FEATURES USED IN OUR XG MODEL

Variable	Description
distance to goal	Distance between the shooter and the middle of the goal line, normalized to StatsBomb pitch size of 120 × 80 due to varying pitch sizes.
shot angle	By creating a triangle between the shooter and the two goalposts, the shot angle is the angle formed at the shooter's position between lines to each post.
distance to the nearest defenders	By measuring the straight line from the shooter to the closest opposing field player, this is how much space the shooter has before a defender can block.
angle to the nearest defenders	By creating a triangle between the shooter, the defender, and the center of the goal, this is the angle at which the defender is positioned relative to the shooting path.
distance to goalkeeper	By measuring the straight line from the shooter to the goalkeeper.
angle to goalkeeper	By creating a triangle between the shooter, goalkeeper, and the center of the goal.

C. xG Model Training and Validation

To estimate the probability of a shot resulting in a goal (xG), we trained a supervised classification model using the XGBoost algorithm. The dataset was constructed from StatsBomb event data and included only valid shooting events with complete information on spatial and contextual features.

Due to the strong class imbalance in shot-to-goal data, we used a stratified 5-fold cross-validation strategy to maintain the original class proportions across all training and validation splits. Instead of applying explicit class weighting (such as `scale_pos_weight`), we relied on comprehensive hyperparameter tuning.

The XGBoost model was trained using a grid search over a range of hyperparameters, the performance of the model was evaluated using ROC-AUC, recall and precision, which is a robust performance metric for imbalanced binary classification tasks. Once the best configuration was identified, the model was refitted on the entire dataset. The calculated probabilities were then compared to StatsBomb xG values to assess overall consistency. To evaluate the reliability of the model outputs as probability estimates, we computed additional metrics including Mean Squared Error (MSE), Brier Score, and a calibration curve.

These calculated xG values will serve as the basis for the normalized xG-derived metrics used in the subsequent evaluation of players, goalkeepers, teams and coaches throughout this paper (Figure 1).

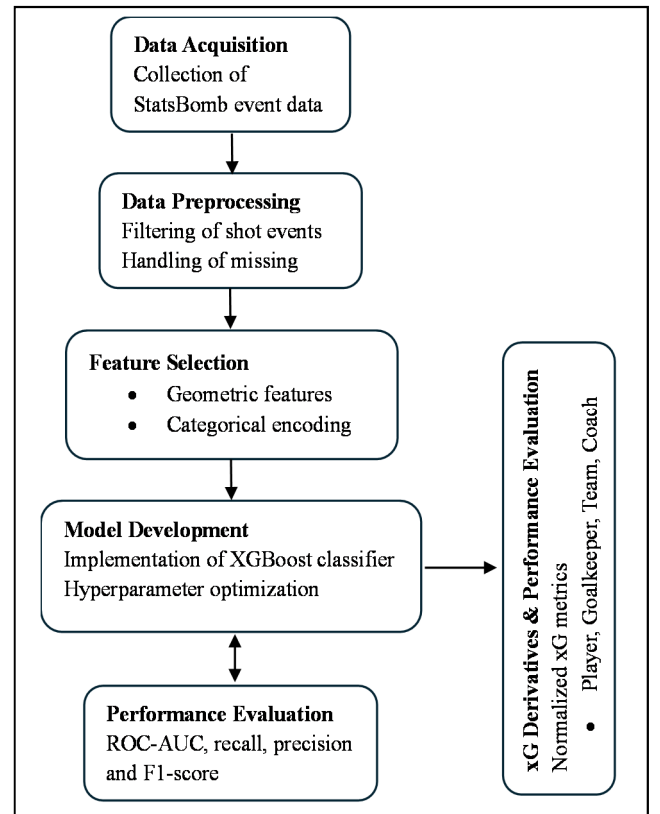


Fig. 1. Flowchart of the proposed xG modeling methodology

V. RESULTS AND DISCUSSION

Our comprehensive analysis of xG metrics derived from the XGBoost model reveals several key dimensions of performance in football:

A. Model Performance

The performance of the xG prediction model, developed using the XGBoost algorithm, was evaluated on the full dataset following hyperparameter tuning via stratified 5-fold cross-validation. The final model, optimized with the best hyperparameters (learning_rate=0.1, max_depth=3, n_estimators=50, subsample=0.8, colsample_bytree=0.8), achieved a ROC-AUC score of 0.8574, indicating a strong discriminative capacity between goal and non-goal outcomes.

Despite the high-class imbalance (only 427 goals among 5107 shots), the model showed robust performance. It successfully identified 66% of actual goals (recall = 0.66) and yielded a precision of 0.33, meaning that one out of three predicted goals were correct, this is a typical result in imbalanced classification tasks. The model achieved an overall accuracy of 86%, with a macro F1-score of 0.68 and a weighted F1-score of 0.88, reflecting consistent performance across both classes.

In terms of probability calibration, the Mean Squared Error (MSE) between the calculated xG value and the StatsBomb xG reference value was 0.0058, while the Brier score was 0.0593, indicating well-calibrated predictions. The calibration curve (Figure 2) shows a generally good alignment between predicted probabilities and actual outcomes, especially in the low-to-mid probability range. Minor overestimations appear in higher ranges, but overall, the model demonstrates consistent and reliable probability estimates.

Overall, these results confirm the model's effectiveness in estimating shot quality and support the use of its calculated xG values in further analyses of player, goalkeeper, team, and coach performance.

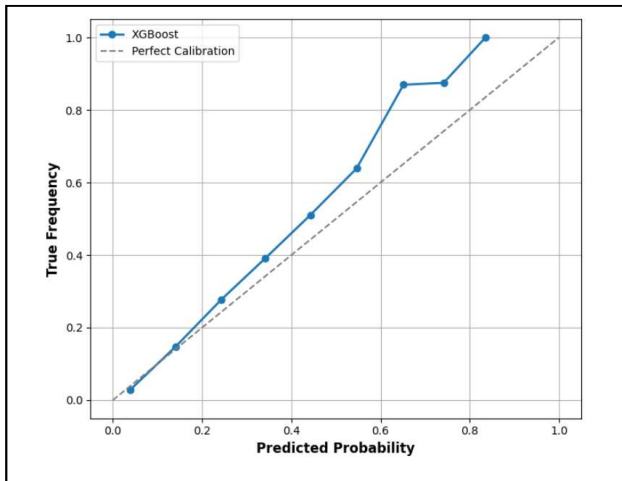


Fig. 2. The Calibration Curve.

B. Player and Goalkeeper Evaluation

As represented in Figure 2, an analysis of the Botola Pro1 top 10 shooters revealed significant variations in finish efficiency. Guy Mbenza (15 goals, xG: 8.93, Delta finishing: +0.085) and Reda Hajhouj (Delta finishing: +0.058) exhibited consistently strong performance, converting chances above expected levels. In contrast, Oussama Lamlaoui (Delta finishing: -0.012) and Saad El Morsli (Delta finishing: -0.0004) underperformed marginally relative to their xG, though their deviations were negligible, suggesting near-average efficiency.

These findings underscore the utility of xG and Delta finishing as objective metrics for evaluating finishing quality. Predominantly positive delta affirms strong finishing quality, while negative deviations highlight areas for improvement. This approach provides a more nuanced assessment than raw goal counts, aligning with modern analytical standards in player evaluation.

As shown in Table II, the xGP (Expected Goals Prevented) metric highlights meaningful differences in goalkeeper performance. Khalid Kbiri Alaoui achieved the highest score (xGP = 0.34), indicating strong shot-stopping ability relative to the quality of shots faced. He was followed by Aymane Majid (xGP = 0.30) and Mohamed Ferni (xGP = 0.26), both of whom also demonstrated solid defensive contributions by saving a high number of shots and conceding relatively few goals. In contrast, several goalkeepers showed lower xGP values (under 0.20), suggesting difficulties in handling high-quality shooting situations.

This variation in xGP scores confirms the usefulness of this metric for evaluating goalkeeper efficiency beyond simple saving counts. It reflects how well a goalkeeper performs compared to the difficulty of the shots faced. Figure 3 illustrates this relationship visually, using a bubble chart where goalkeepers are positioned based on shots saved and shots conceded, with bubble size and color indicating xGP values. This visualization helps identify the most effective goalkeepers in the Botola Pro1.

TABLE II. GOALKEEPER EVALUATION WITH EXPECTED GOALS PREVENTED (xGP).

goalkeeper_name	Shots Saved	Goals	xGP
Khalid Kbiri Alaoui	31	5	0.3362
Aymane Majid	67	12	0.3009
Mohamed Ferni	45	9	0.2630
Mokhtar Majid	86	20	0.2624
Mehdi Maftah	90	23	0.2502

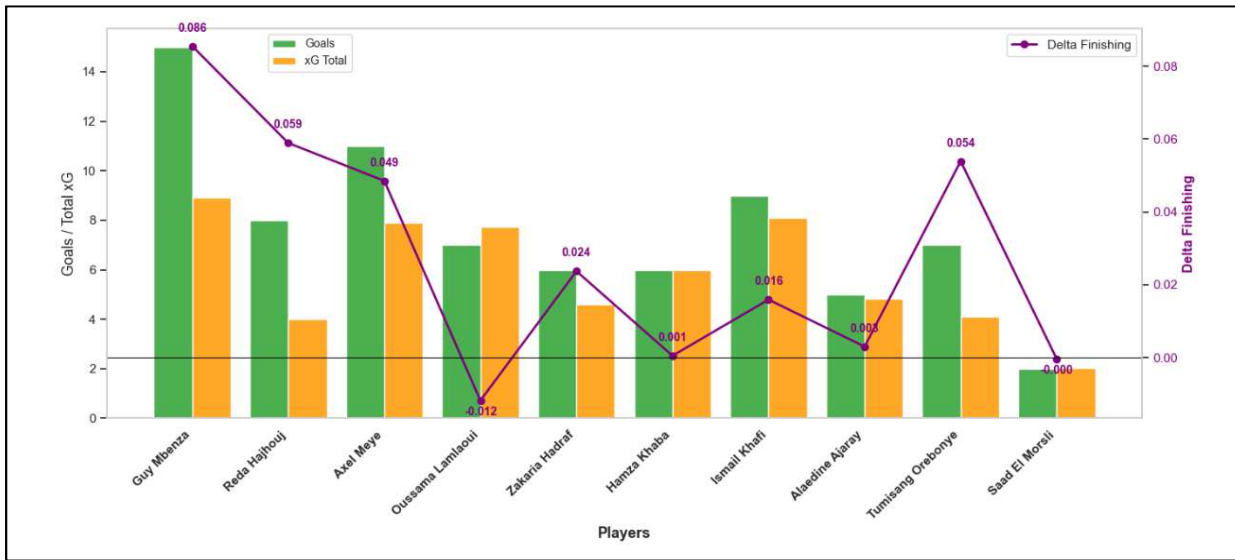


Fig. 3. Comparison of predicted xG values and actual goals scored, showing how xG captures chance quality beyond goal counts.

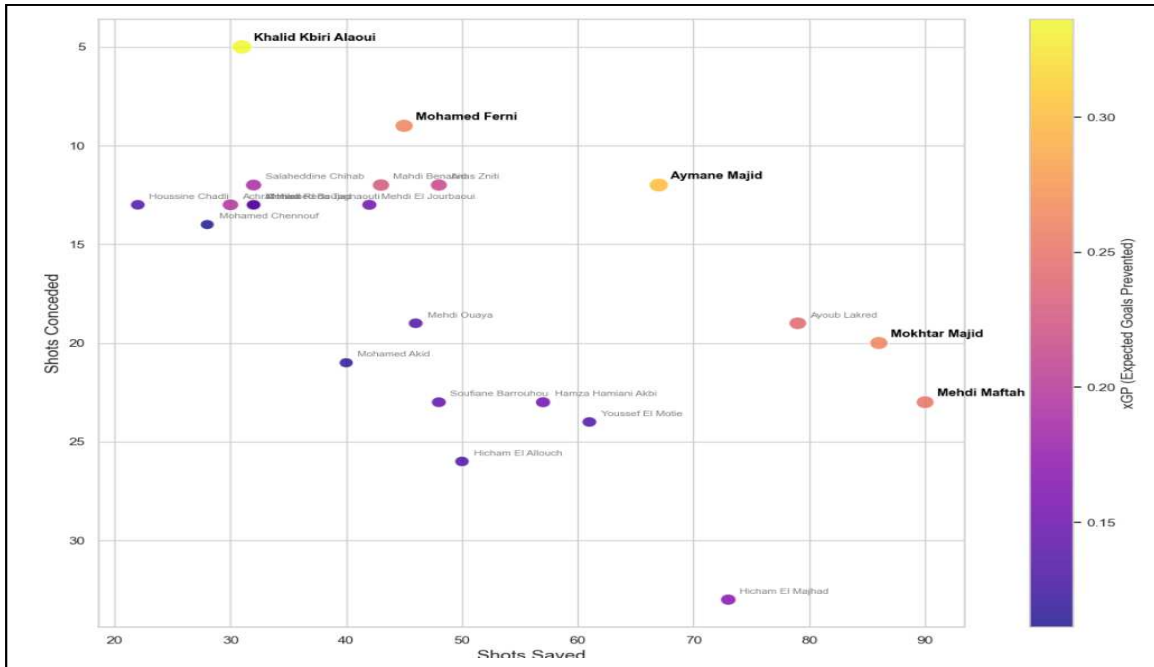


Fig. 4. Distribution of Goalkeepers Based on Shots Saved vs. Shots Conceded, Colored by Expected Goals Prevented (xGP).

C. Team Performance Metrics

Net xG difference highlighted Raja Casablanca (+0.06) and Wydad Casablanca (+0.04) as the most balanced teams, reflecting dominance in both chance creation and defensive solidity. However, when adjusting for conversion rates via Team xG Efficiency, even top teams displayed negative values (e.g., FAR Rabat: -0.31), suggesting systemic

inefficiencies in finishing across the league, while defensively, Maghreb Fes (+0.32) and Olympic Safi (+0.34) exceeded expectations. These contrasting trends across teams are visually represented in Figure 4, which presents the Net xG Difference (Offensive – Defensive) values per team, offering a visual summary of each clubs.

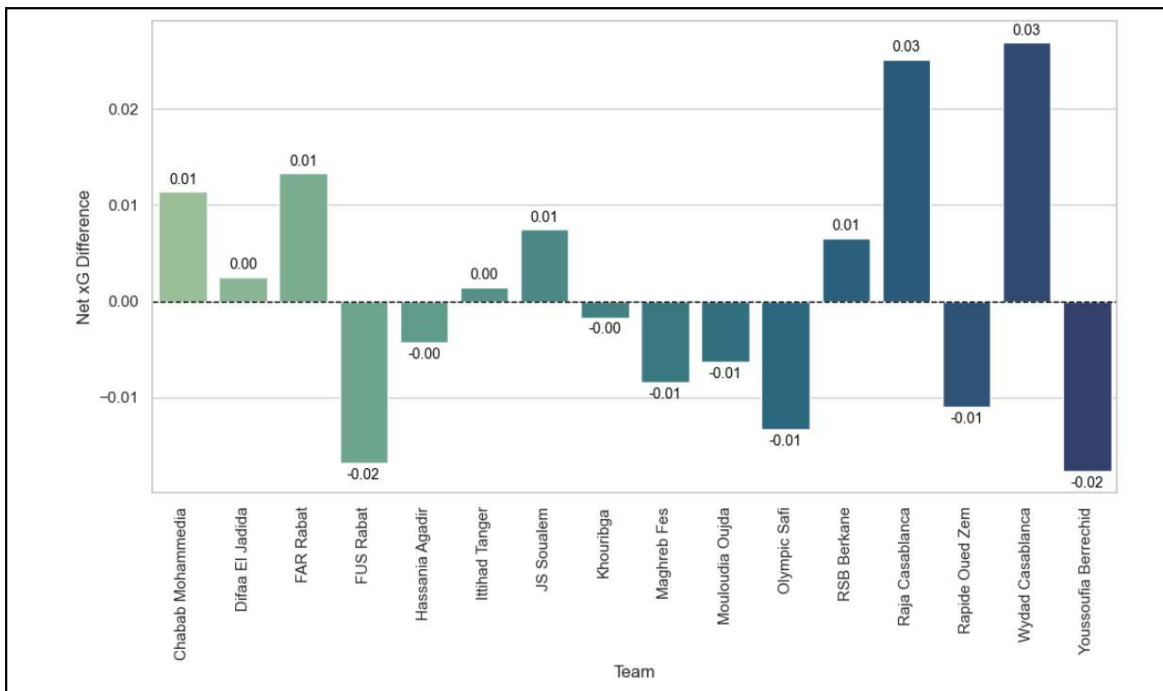


Fig. 5. Net xG Difference by team, where positive values indicate attacking dominance and negative values reflect defensive weakness.

D. Coach Evaluation

Opponent-Adjusted xG (OA-xG) analysis revealed that Wydad Casablanca (+0.0218) and Raja Casablanca (+0.0203) had the highest net OA-xG values, suggesting that their coaches were particularly effective at preparing their teams to create high-quality chances even against defensively strong opponents. The xGTA metric provides insight into teams' ability to generate or concede goal-scoring opportunities across match intervals. The most notable positive peaks occurred for JS Soualem in the 45–60' period (+2.1) and for RS Berkane in the first 15 minutes (+2.4), highlighting strong offensive momentum during key phases. Chabab Mohammedia and Hassania Agadir also showed sharp increases after halftime, with values above +1.5, suggesting effective tactical readjustments. Conversely, Olympic Safi and Rapide Oued Zem experienced severe drops in the final 15 minutes (–2.5 and –2.7, respectively), indicating late defensive collapse. Teams like Khouribga and FUS Rabat displayed volatile performance, alternating between peaks and troughs, while others such as Raja Casablanca and Wydad Casablanca maintained more balanced xGTA curves, reflecting steady control. Overall, the xGTA analysis highlights significant intra-match tactical disparities, particularly during the 45–60' and 75–90' intervals. These variations demonstrate how tactical decisions, physical readiness, and substitution strategies can significantly influence the net expected goals during critical match phases.

To check if our xG model gives reliable results, we used the same method of evaluation with the official StatsBomb xG values from the dataset. The results we got from our model and StatsBomb are very close in all categories: players, goalkeepers, teams, and coaches.

For the top goal scorers, names like Guy Mbenza, Reda Hajhouj, and Axel Meyer appear at the top in both lists. Even if the total xG is slightly different (for example, Mbenza has 8.93 with our model and around 9.83 with StatsBomb), yet the ranking and underperformance conclusion remained identical.

This shows that both methods lead to the same findings about player performance. In the goalkeeper ranking, Khalid Kbiri Alaoui, Aymane Majid, Mokhtar Majid and Mehdi Maftah keep their places with very small differences in values. For example, Khalid Kbiri Alaoui has an xGP of 0.336 with our model and 0.323 with StatsBomb. At the team level, clubs such as Raja Casablanca and Wydad Casablanca are ranked highest in both Net xG and xG Efficiency. Teams like Maghreb Fès and Olympic Safi also show good defensive performance with both xG types.

Finally, for coaching performance, Wydad Casablanca and Raja Casablanca appeared as leaders using both calculated xG and StatsBomb xG in Opponent Adjusted xG analysis, which means tactical quality was captured at the same regardless of the xG source.

In general, these results show that our calculated xG is very close to the expert StatsBomb data. This gives more confidence in our model and shows that it can be used for fair and accurate performance analysis. Our analysis shows that xG metrics give us detailed ways to measure performance at every level of football. The model works well at identifying good scoring chances, and the different xG measures we created help coaches and players make better decisions. These tools can improve training, game strategies, and team management. We also found some areas we can make even better, like making the probability calculations more precise and including more game situations.

Beyond descriptive statistics, the proposed indicators also carry tactical meaning. The variations in xGTA values, for example, show that substitutions and specific match periods can strongly influence team performance. Moroccan clubs displayed greater fluctuations in finishing and defensive balance compared to European benchmarks, which may reflect differences in tactical discipline and physical preparation. This comparison suggests that our framework is useful not only for evaluating local football but also for

positioning Moroccan teams within an international benchmarking perspective.

VI. CONCLUSION

This study proposed a practical and event data-driven for evaluating football performance in Morocco using normalized xG-derived metrics. By combining contextual features with a tuned XGBoost model, we introduced reliable indicators to assess players, goalkeepers, teams, and coaches. The results confirmed that efficiency-based metrics help reveal underperformance trends, especially in finishing and defensive phases. Our analysis also highlighted match-time variations and tactical differences across teams. This approach provides clubs and analysts with effective tools and deeper insights into performance, helping them make better decisions in training, tactics, and recruitment. In the long term, it can contribute to raising the overall level of Moroccan football.

However, this work has some limitations. The analysis relies on data from a single season of Botola Pro 1, which may limit generalization to other contexts. In addition, the class imbalance between goals and non-goals introduces challenges for prediction accuracy. Despite these constraints, the framework demonstrates strong potential for local and international application.

Future work will focus on extending the dataset to multiple seasons, applying advanced balancing techniques, and integrating new indicators related to ball possession and tactical phases. These improvements will strengthen the accuracy and practical relevance of the proposed framework.

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